



Collection Date:

FINAL DIAGNOSIS:

THYROID, TOTAL THYROIDECTOMY (34 GRAMS) -

A. PAPILLARY THYROID MICROCARCINOMA, (0.5 CM) ONCOCYTIC/WARTHIN-LIKE VARIANT, CONFINED TO LEFT LOBE.

- I. TUMOR IS SUBCAPSULAR
- II. PSAMMOMA BODIES IN LYMPHATICS (CONSIDERED EQUIVALENT OF ANGIOLYMPHATIC INVASION AND INTRAGLANDULAR SPREAD)
- III. NO SIGNIFICANT FIBROSIS

B. FOUR (4) ISTHMIC LYMPH NODES, NO TUMOR PRESENT (0/4).

C. PATHOLOGIC STAGE: pT1a N0.

D. NODULAR THYROID HYPERPLASIA WITH DOMINANT ONCOCYTIC (HURTHLE CELL) NODULE (1.0 CM) AND EXUBERANT CHRONIC LYMPHOCYTIC THYROIDITIS.

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE: Total Thyroidectomy
TUMOR SITE: Left Lobe
TUMOR FOCALITY: Unifocal
TUMOR SIZE (largest nodule): Greatest Dimension: 0.5 cm
HISTOLOGIC TYPE: Papillary carcinoma
PRIMARY TUMOR (pT): pT1a
REGIONAL LYMPH NODES (pN): pN0

Number of regional lymph nodes examined: 4

Number of regional lymph nodes involved: 0

DISTANT METASTASIS (pM):

Not applicable

EXTRATHYROIDAL EXTENSION:

Not identified

MARGINS:

Margins uninvolved by carcinoma

LYMPH-VASCULAR INVASION:

Present

ADDITIONAL PATHOLOGIC FINDINGS:

Thyroiditis

Comment:

Psammoma bodies in lymphatics considered positive for angiolymphatic invasion.

SPECIAL PROCEDURES:

Molecular Anatomic Pathology Procedure

Interpretation

MOLECULAR ANATOMIC PATHOLOGY TESTING:

Block B:

Mutations in *BRAF*, *NRAS*61, *HRAS*61, *KRAS*12/13 and *RET/PTC1* and *RET/PTC3* rearrangements NOT identified.

NOTE:

Nucleic acids were extracted in the amount sufficient for testing.

BACKGROUND:

Mutations in either *BRAF* or *RAS* genes or *RET/PTC* rearrangements are found in more than 70% of papillary thyroid carcinomas (1). Mutations in the *RAS* genes or *PAX8/PPARγ* rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hurthle cell) carcinomas (2). Finding of *BRAF*, *RET/PTC* or *PAX8/PPARγ* mutations in thyroid FNA samples correlates with ~100% probability of cancer and finding of *RAS* mutation correlates with malignancy in 74-85% of cases (3-5). Based on our prospective study at risk of cancer in thyroid nodules negative for all of these mutations is 5.9% in nodules with atypia of undetermined significance/follicular lesion of undetermined significance (AUS/FLUS) cytology (including the 2.3% risk of invasive cancer and 0.6% risk of cancer with extrathyroidal spread in this group), 14% in nodules with follicular neoplasm/suspicious for follicular neoplasm (FN/SFN) cytology, and 28% in nodules with suspicious for malignant cells (SMC) cytology (5). In addition, *BRAF* V600E (T1799A) mutation has been associated with more aggressive behavior of papillary carcinoma (6) and may be used for risk stratification of patients with papillary thyroid cancer (7). However, single cases of false-positive *BRAF* mutation detection in thyroid FNA samples have been reported in the literature (8); therefore, the results of molecular testing should be interpreted in combination with clinical information, imaging, and cytology.

1. Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. *Am J Surg Pathol* 2006; 30:216-222.
2. Nikiforova MN, et al. RAS point mutations and PAX8-PPARγ rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. *J Clin Endocrinol Metab* 2003; 88: 2318-28.
3. Nikiforov YE et al. Molecular testing for mutations in improving the fine-needle aspiration diagnosis of thyroid nodules. *J Clin Endocrinol Metab*. 2009 Jun;94(6):2092-8.
4. Cantare S, et al. Impact of proto-oncogene mutation detection in cytological specimens from thyroid nodules improves the diagnostic accuracy of cytology. *J Clin Endocrinol Metab* 2010;95:1365-1369.
5. Nikiforov YE, et al. Impact of Mutational Testing on the Diagnosis and Management of Patients with Cytologically Indeterminate Thyroid Nodules: A Prospective Analysis of 1056 FNA Samples. *J Clin Endocrinol Metab*. 2011 96(11):3390-7.
6. Elisei R, et al. *BRAF* V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. *J Clin Endocrinol Metab* 2008; 93:3943-3949.
7. Xing M, et al. *BRAF* mutation testing of thyroid fine-needle aspiration biopsy specimens for preoperative risk stratification in papillary thyroid cancer. *J Clin Oncology* 2010; 27:2977-82.
8. Chung KW, et al. Detection of *BRAF*V600E mutation on fine needle aspiration specimens of thyroid nodule refines cyto-pathology diagnosis, especially in *BRAF*V600E mutation-prevalent area. *Clin Endocrinol (Oxf)*. 2008; 65(5):680-8.

MUTATIONAL ANALYSIS:

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the LightCycler platform to amplify *BRAF* codons 599-601, *NRAS* codon 61, *HRAS* codon 61, and *KRAS* codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. Single-step real-time RT-PCR was performed to amplify the fusion points of *RET/PTC1* and *RET/PTC3* rearrangements on ABI7500. If required, the mutation type was confirmed by Sanger sequencing of the PCR products on ABI3130. Amplification of the *KRT7* genes was used to control quality and quantity of RNA and amount of epithelial cells present in the specimen. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal DNA and RNA.

Criteria	Yes	No
Diagnosis Discrepancy		
Primary Tumor Site Discrepancy		
HIPAA Discrepancy		
Prior Malignancy History		
Dual/Synchronous Primary Noted		
Case is (circle): QUALIFIED / DISQUALIFIED		
Reviewer Initials: <i>PLT</i> Date Reviewed: <i>8/2/2012</i>		

8/2/12